INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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Data-Driven Optimization of Hospitality Operations:

A Case Study of Hotel SR Paark, Kavali

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Mid Term Submission for the Project on Business Data Management

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Declaration Statement

I am working on a Project Title "**Data-Driven Optimization of Hospitality Operations: A Case Study of Hotel SR Paark, Kavali**". I extend my appreciation to **Hotel SR Paark**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate

Name: Mannepalli Bala Praharsha

Date: 04-01-2026

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1. Executive Summary

Hotel SR Paark is a B2C hospitality company within the vicinity of APSRTC Bus Stand in Kavali, Andhra Pradesh with one suite, twenty-two deluxe rooms and business based banquet services to business travelling families and group visitors. Despite the hotel being in demand in the weekend and peak seasons, there are consistent issues that include low occupancy on weekdays, the inconsistent use of rooms and fluctuating income that directly impact profitability and efficiency of operations.

The physical booking registers of the hotel in the month of January 2025 to December 2025 were used as the primary data and were digitized in the form of a MS Excel spreadsheet. The dataset includes key metadata variables such as Date, Rooms Occupied, Total Revenue, Extra Bed Count, Cold Drinks Count, Stay Type, and Remarks. The descriptive statistics show that the average occupancy rate is 5.52 rooms/day, and the revenue ranges between ₹2,300 and ₹29,000, which means that the demand is very variable. The use of add-on services also brings out the upsell prospects.

The analysis was carried out using Microsoft Excel over a period of two months (November–December 2025). Section-3 presents metadata definitions and descriptive statistics, Section-4 explains trend analysis, regression and price-elasticity based methods, while Section-5 summarizes the key findings including seasonal demand imbalance, revenue volatility and under-utilization of add-on services. Mid-term analysis revealed occupancy below 50% capacity during Jan–Mar and revenue volatility of ₹7,331 indicating unstable cash flows.

2. Proof of Originality

As a part of the evidence of the originality of the obtained material, I will be offering the following supporting evidence and resources: a letter of the organization, the photograph of the firm/servicescape, and a video recording of an interaction with the hotel manager. With these materials, they are safely fixed with the help of the following Google Drive link to be referred and reviewed:

https://drive.google.com/drive/folders/1jZXI0MIaGRJ8w0LW_nUbUDHt2y4vQqKN?usp=drive_link

The video communication mode is mostly in Telugu with English and an English version of the interaction has been shown in the same Google drive in case of academic reference. The same Drive folder also has photographs of the hotel premises, entrance board, reception area, banquet hall and booking registers. To confirm the primary data collection process, the No Objection Certificate (NOC) by Hotel SR Paark on official letterhead with signature and stamp that allows the use of the booking data in academic purposes is supplied in the folder. All these materials make the information relied upon in this project credible and original.

3.1. Meta Data

In this project, data were collected systematically between 1 st January 2025 and 31 st December 2025 and these data included the daily room occupancy and revenue information at Hotel SR Paark, Kavali. This information allows determining the true performance of the hotel regarding its operations and revenue trends throughout the year. The physical booking registers of the hotel were copied manually to the MS Excel and analyzed there.

The hotel adheres to a manual system of registering books, whereby the particulars of the room that has been used, the revenue that has been generated on a daily basis and additional services of the room are registered rather than the customer details, thus privacy is ensured. The obtained data were arranged in the following columns:

- **Date:** the calendar date of room booking activity is noted in the form of DD/MM/YYYY so that these can be analysed chronologically and help in the identification of seasonal trends.
- **Rooms Occupied:** This is a numerical column, which the hotel uses to capture the number of rooms occupied per day, which is a reflection of the daily occupancy rates.
- **Total Revenue (₹):** Shows the cumulative value of money taken in daily room bookings and services, which is the main indicator of profitability.
- **Extra Bed Count:** It is the number of extra beds ordered on a given day, which indicates upselling behaviour and additional revenue.
- **Cold Drinks Count:** It records the number of cold drinks sold per day, which means more service usage.
- **Stay Type:** A categorical column that indicates the nature of stay such as short stay and full-day stay, which helps in the assessment of pricing strategy.
- **Remarks:** Exceptions to bulk bookings or event stays.

The data analysis was done through Microsoft Excel where formula, pivot table and charts were used to reveal the occupancy pattern, revenue pattern and the behaviour of utilising the services.

Date	Rooms_Occupied	Total_Revenue	Extra_Bed_Count	Cold_Drinks_Count	Stay_Type	Remarks
11-01-2025	6	14004	2	14	Business	
12-01-2025	2	6500	0	6	Business	Regular
13-01-2025	1	2300	0	3	Solo	Regular
14-01-2025	9	21300	3	22	Group	
15-01-2025	9	29000	4	25	Family	2-day stay
16-01-2025	7	11600	2	16	Business	Regular
17-01-2025	10	19102	3	26	Business	
18-01-2025	4	12300	1	10	Family	Regular
19-01-2025	7	22400	2	18	Family	
20-01-2025	4	9562	1	11	Group	
21-01-2025	4	11200	1	10	Business	Regular
22-01-2025	6	8500	2	14	Business	Regular
23-01-2025	3	2600	0	6	Solo	
24-01-2025	5	10500	1	12	Group	Function
25-01-2025	2	3300	0	5	Solo	Regular
26-01-2025	1	2800	0	3	Solo	
27-01-2025	8	18738	3	21	Business	
28-01-2025	9	18000	3	24	Family	Regular
29-01-2025	3	7269	1	8	Business	
30-01-2025	8	18502	3	22	Business	
31-01-2025	8	11500	2	19	Group	
AVERAGES	5.523809524	12427.47619	1.619047619	14.04761905		
STANDARD DEVIATION	2.805081233	7154.551729	1.214052265	7.299551131		
SUM		260977	34	295		

Fig 1 : Monthly Summarized Data

3.2. Descriptive Statistics

The following section is the quantitative summary of primary data to be collected within the period of 1 st January 2025 to 31 st December 2025 in Hotel SR Paark, Kavali. The aim is to know the room occupancy behavior, revenue fluctuation, and add-on service utilization patterns that are directly related to the issues of poor occupancy and unpredictable revenue of the hotel.

1. Measures of Central Tendency

Statistic	Rooms Occupied	Total Revenue (₹)	Extra Bed Count	Cold Drinks Count
Mean	5.52	12,427	1.62	14.05
Median	6	11,500	2	14
Mode	7	11,200	2	12

- **Interpretation and Justification:** The median of 6 rooms and average 5.52 rooms/ day proves that the hotel does not use more than half a capacity per day on average, which is an obvious explanation of the issue of under-utilization of the weekdays. The mean of ₹12,427 and a frequency of ₹11,200 shows that the revenue is concentrated in the average profitability and this confirms that the problem of low profitability during low seasons exists. Mean extra beds and cold drinks values prove the existence of add-on services, which are used regularly but not completely, which proves the issue of lost auxiliary funds.

2. Measures of Variability

Statistic	Rooms Occupied	Total Revenue (₹)
Standard Deviation	2.87	7,331
Minimum	1	2,300
Maximum	10	29,000

- The high volatility of occupancy and revenue in terms of high variation of standard deviation 2.87 rooms and ₹7,331 revenue. This gap in minimum and maximum revenue (₹2,300 to ₹29,000) proves the fact of extreme instability of income, which in turn is the direct cause of the issue in the hotel of unstable cash flow and the absence of price-dependent on the demand.

3. Frequency & Pattern Analysis

- The occupancy value of 7 rooms/day is most prevalent with extreme low occupancy values of 1-2 room/day which is experienced during off-season and weekdays. This pattern highlights:
    - Excess reliance on weekends and summer seasons.
    - Low consistency in demand during the weekdays.
    - Lack of systematic pricing or marketing policies at times of low demand.
- **Justification:** These trends are the obvious reasons why dynamic pricing and weekday-offers, as well as optimization techniques related to booking, should be applied to flatten occupancy and stabilize the revenue across the year.

These descriptive statistics quantitatively confirm the existence of inconsistent occupancy, unstable revenue, and under-utilization of add-on services, thereby directly supporting the identified business problems of Hotel SR Paark.

4. Detailed Explanation of Analysis Process

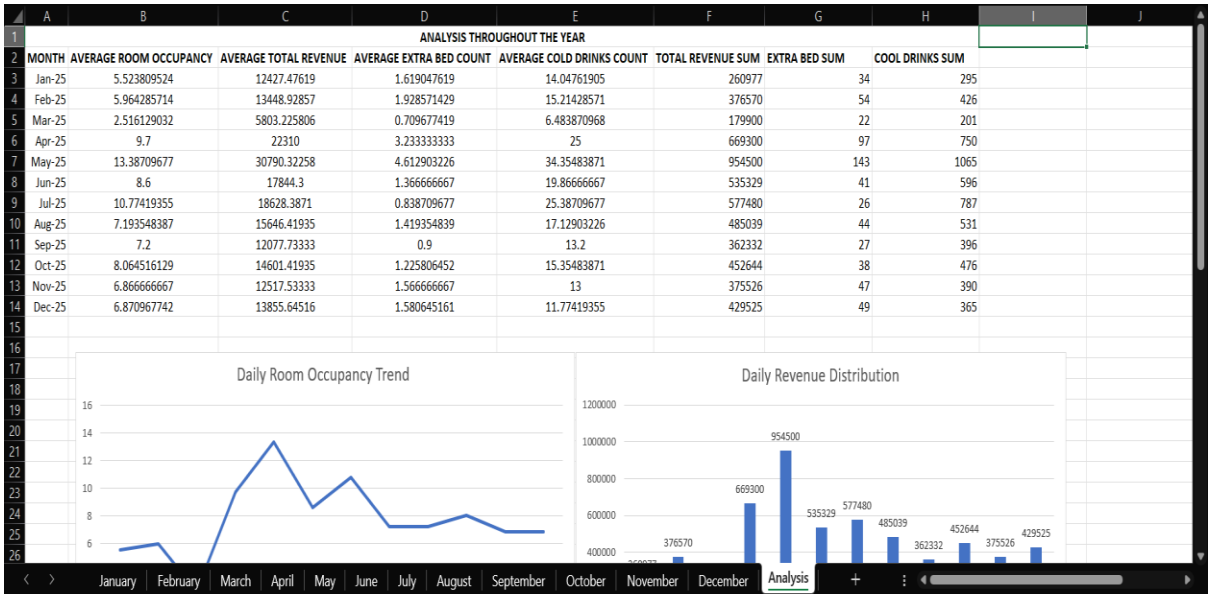
- ## Data Cleaning and Preprocessing
  - The crude data was hand-stated out of the physical booking registers of the hotel and was converted into MS Excel format. In the process, it was found that missing values, irregular date format, and non-numeric revenue items were found. All date fields were changed to a standard DD/MM/YYYY format and columns that included revenue were changed to numeric format. Personal identifiers were removed to ensure privacy, and conditional formatting was used to highlight outliers caused by bulk event bookings.
  - **Importance:** Cleaning of data guarantees accuracy, consistency and reliability of data. The trend analysis and statistical calculations would not yield reliable results without the standardization of the dates and the elimination of the erroneous entries that would consequently influence the reliability of the decision-making.
- ## Analysis Process
  - **Occupancy Trend Analysis:**
    - Line charts were relied on to determine the patterns of fluctuation and demand of room occupancy on a daily basis.
    - Justification: Line charts were used to identify seasonal trends and weekday–weekend variations in daily room occupancy.
  - **Revenue Pattern Analysis:**
    - Daily revenue was analyzed using bar charts to identify peak earning days and periods of low performance.
    - Justification: This analysis helps understand revenue fluctuations over time and highlights days with unusually high or low income for better financial planning.
  - **Occupancy-Revenue Relationship:**
    - Excel was used to perform a simple linear regression to analyze the relationship between room occupancy and total revenue.
    - Justification: This method quantifies how strongly revenue depends on occupancy levels and helps assess the extent to which changes in occupancy drive revenue variation.
  - **Dynamic Pricing Feasibility Assessment:**
    - Elasticity of demand was calculated in order to know how sensitive customers are to price.

- Justification: This approach favours the development of dynamic pricing policies.

- **Upselling Opportunity Analysis:**
  - The influence of additional bed and sales of cold drinks on the revenue was investigated with the help of column charts.
  - Justification: The analysis of auxiliary services reveals more sources of revenue

These analytical measures directly tackle the main issue of erratic occupancy and unchartable revenue development because of setting the basis of data-supported strategies of optimization in booking and revenue control.

Summarized Dataset & Analysed Sheet Link:
https://docs.google.com/spreadsheets/d/1YeWQH9Ebeft8fXTmU-YK7m8h8BuB9lFw/edit?usp=sharing&ouid=111381051408798734177&rtpof=true&sd=true

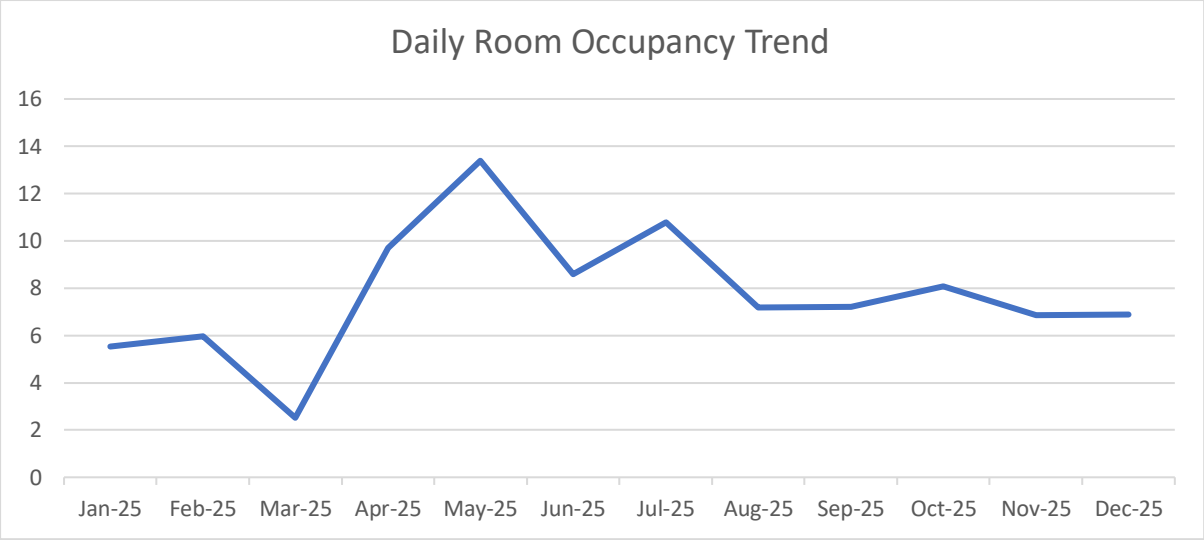


MONTH	AVERAGE ROOM OCCUPANCY	AVERAGE TOTAL REVENUE	AVERAGE EXTRA BED COUNT	AVERAGE COLD DRINKS COUNT	TOTAL REVENUE SUM	EXTRA BED SUM	COOL DRINKS SUM
ANALYSIS THROUGHOUT THE YEAR							
Jan-25	5.523809524	12427.47619	1.619047619	14.04761905	260977	34	295
Feb-25	5.964285714	13448.92857	1.928571429	15.21428571	376570	54	426
Mar-25	2.516129032	5803.225806	0.709677419	6.483870968	179900	22	201
Apr-25	9.7	22310	3.233333333	25	669300	97	750
May-25	13.38709677	30790.32258	4.612903226	34.35483871	954500	143	1065
Jun-25	8.6	17844.3	1.366666667	19.86666667	535329	41	596
Jul-25	10.77419355	18628.3871	0.838709677	25.38709677	577480	26	787
Aug-25	7.193548387	15646.41935	1.419354839	17.12903226	485039	44	531
Sep-25	7.2	12077.73333	0.9	13.2	362332	27	396
Oct-25	8.064516129	14601.41935	1.225806452	15.35483871	452644	38	476
Nov-25	6.866666667	12517.53333	1.566666667	13	375526	47	390
Dec-25	6.870967742	13855.64516	1.580645161	11.77419355	429525	49	365

Fig 2 : Analyzed sheet

5. Results and Findings

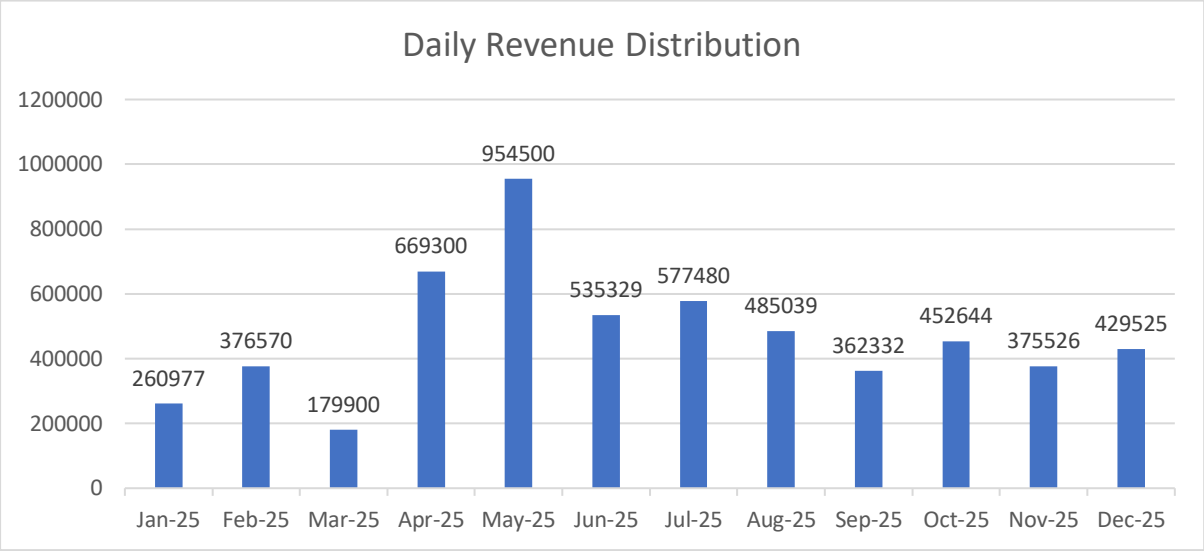
5.1. Daily Room Occupancy Trend



The line plot indicates that there is a high seasonality in terms of room booking. The occupancy is low in the months of January to March, and the lowest in March (≈2-3 rooms), which means that the occupancy is underutilized during the off-season. The occupancy then suddenly increases between April and May peaking in May (≈13-14 rooms) equivalent to the summer vacation and the local event seasons. Occupancy decreases slowly after May but stabilizes at 7-8 rooms per day between August and December which means that the demand is moderate but consistent.

- **Insight:** The hotel experiences strong seasonal imbalance in demand, with severe under-utilization during the first quarter of the year and heavy pressure during the summer months. This uneven pattern indicates inefficient capacity usage and highlights the need for strategies to stabilize occupancy across different seasons and weekdays.

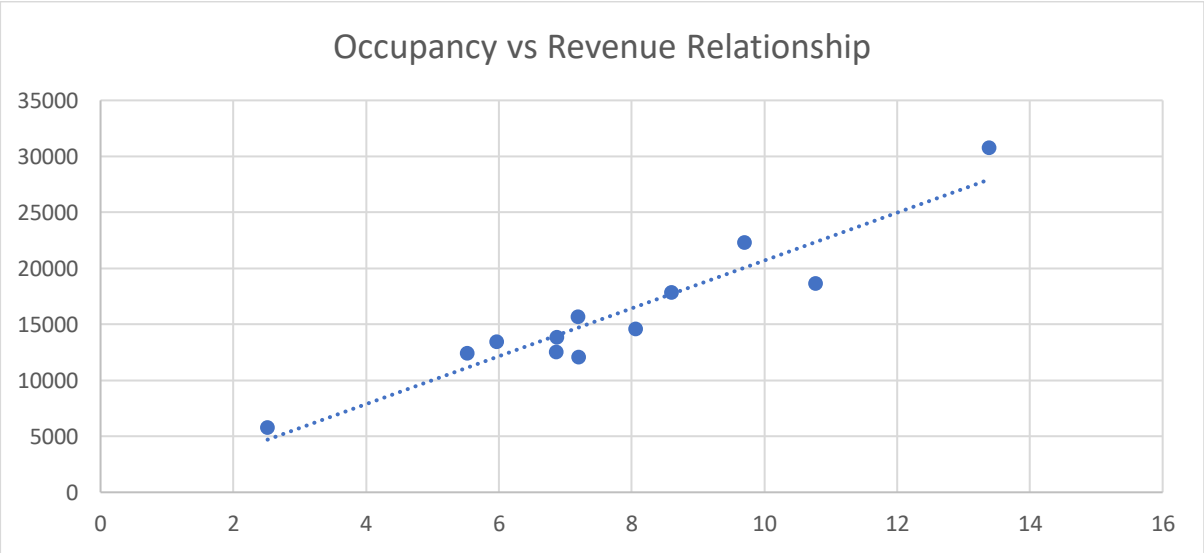
5.2. Seasonal Revenue Variation



The occupancy continues to reflect in the revenue bar chart. The lowest revenue is in March (₹1.79 lakhs) and the highest during the month of May (₹9.54 lakhs). Peak occupancy months are April, May, June and July, which are high revenue months. The decline in revenue subsequently follows in the period after August, although the occupancy has not declined yet, which shows that optimized pricing mechanisms are not in operation.

- **Insight:** The post peak season growth of revenue is not proportionate, and that indicates the missed price optimization opportunities.

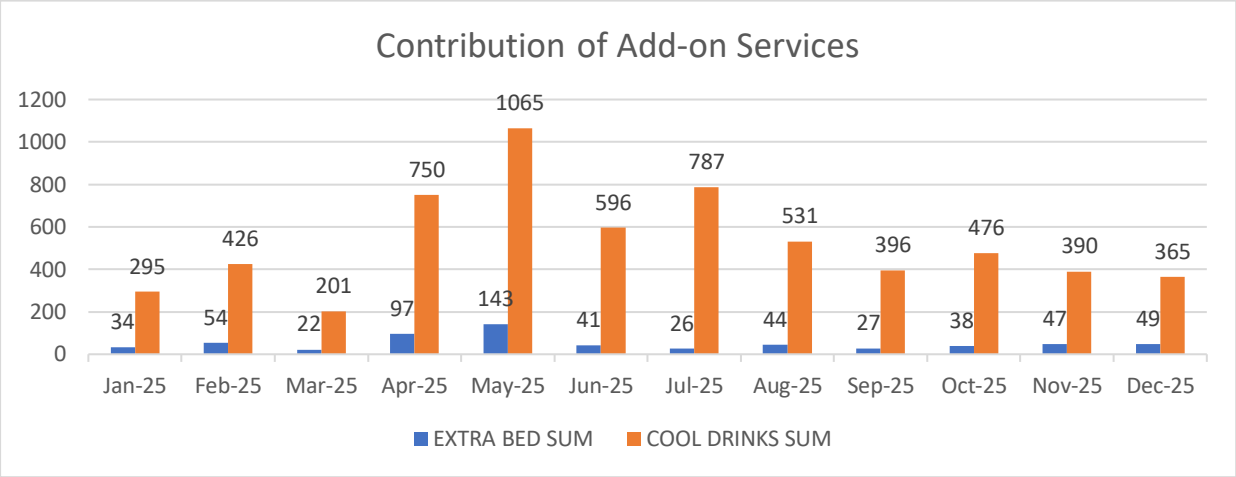
5.3. Occupancy vs Revenue Relationship



The trend line of the scatter plot indicates that there is a strong positive linear relationship between the rooms occupied and the total revenue. The higher the occupancy between 2 to 14 rooms, the higher the revenue which soars exponentially between ₹5,000 to ₹30000 per day.

- **Insight:** The revenue is mostly dependent on the occupancy of the room thus proving that the best way of raising the profitability is through enhancement of consistency of occupancy.

5.4. Contribution of Add on Services



This chart shows that cold drinks have the highest add-on volume and this is topped with a clear peak in the month of May (1065 units). The sales of cold drinks in the period between April and July are consistently high and this directly relates to the high rate of hotel occupancy in the peak season.

The trend of more extra beds during the months is also observable in the months of April, May, October, and December which shows that there are more group and family stays during holiday and vacation seasons.Generally, the trend has shown clearly that the consumption of add-on services is directly dependent on the occupancy patterns. The utilization of add-on services is also increased to a higher level when the demand of rooms is high.

- **Insight:** Add-on services will be a powerful revenue generator in peak seasons and an unexplored profit source in low-occupancy months. The overall hotel revenues can be stabilized through strategic promotion of such services throughout the year.

In short, Hotel SR Paark depends on room occupancy as the main source of revenue, with the highest revenues being received during the summer season, especially in the months of May, when occupancy and utilization of add-on services are at their maximum. The months between January and March stand out as the most weak of the months as room utilization and revenue are considerably low, showing that performance is very poor during the off-season. Even though the occupancy level becomes stable during August-December, the related increase in the revenue is low, which points to the lack of successful demand-based pricing practices. Add-on services, like additional beds and cold drinks have a high contribution when the number of reservations is high in the months and little use when the reservations are low thus missing the auxiliary revenue opportunities. All these findings show us that it is necessary to organize booking optimization, to stimulate demand in weekdays and to create dynamic pricing models that will guarantee the stable revenue all year round and higher profits.